Coursework 2 Brief (Group)

Overview

This coursework series is designed to develop your practical skills in machine learning through structured, comparative experiments on a selected benchmark dataset using **scikit-learn**. You will demonstrate your understanding of dataset investigation, analysis design, performance evaluation, and overfitting prevention while focussing on **experiment design and justification rather than raw accuracy**.

In Coursework 2 you work as an engineering team: frame a question, set the constraints a deployed model must meet, and recommend the pipeline that meets them.

This coursework series prioritises:

- Rigorous **experiment design**
- Clear **justification of modelling choices**
- Meaningful **performance analysis** over raw accuracy

Task

As a group, working on your shared dataset:

1. **Frame an engineering question** and one or two realistic **constraints** a deployed model must meet, justified from the application (e.g. for classification, "it must catch at least 80% of the positive cases"; for regression, "its prediction must be within 10% of the true value for at least 90% of cases").
2. **Choose at least three scikit-learn models** suited to the question, and compare them with a linear or logistic regression reference, as in CW1. Any model family may be in the pool, linear models included; ensemble methods are a common choice, but the pool should follow from your question and constraints.
3. **Compare them under one protocol**: the same split, folds and metric for every model, each tuned first, with training scores beside validation scores, and training and prediction times reported with the runtime they were measured on.
4. **Recommend a pipeline** under your constraints, score it on the test set once, and justify it from the evidence.

There is **no template notebook**.

Weighting

1. **20%** Group report
2. **20%** Oral presentation

Submission

One submission per group.

1. The **report** (maximum 8 pages plus a 1-page appendix) follows the **provided template**, with a statement of contribution at the end (not counted).
2. Your group gives a 10-minute **presentation**, followed by 10 minutes of Q&A.
    - Succinctly communicate your question, constraints and comparative findings (accuracy, computational efficiency, overfitting handling).
    - Justify and clearly explain your recommended pipeline.

    - Answer questions to demonstrate understanding. Questions may be put to any member.
    - To demonstrate teamwork, you do not need to take turns speaking, but your presentation must clearly explain the distinctive contributions made by each member.
3. The **notebook**: one group Jupyter notebook that reproduces every result in the report. It is not marked separately, but is used to check results and contributions. If you work in a repository (e.g. GitHub), give its link on the report's first page.

Contributions

How you divide the work is up to you; for a group of five, one possible split is three model owners, one validation owner and one owner of testing and cost measurement, with the background and writing shared. **The group mark is the default**: an individual's mark may be adjusted where the CW2 peer evaluation on Moodle, the statement of contribution and the Q&A together show a markedly smaller or larger contribution.

Use of AI tools

You may use AI tools as a tutor or coding assistant. Say in the statement of contribution which tools you used and what for. In the Q&A, **any member may be asked to explain any part of the notebook**.

Submission Format

- Group submissions need to be titled
    - "ENG4200_CW2_report_<group ID>.pdf";
    - "ENG4200_CW2_presentation_<group ID>.pdf";
    - "ENG4200_CW2_notebook_<group ID>.ipynb".
- ENG5338 groups use ENG5338 in place of ENG4200.

Assessment Criteria

Report

Grade Band	1. Design of experiments and pipelines (25%)	2. Comparative Analysis of Accuracy and Overfitting Handling (25%)	3. Comparative Analysis of Computational Efficiency (25%)	4. Justification of Suggested Pipeline (25%)
A Excellent	Comprehensive and well-structured experimental design. The engineering question and constraints are realistic and clearly justified, and the pool of models follows from them. Pipelines are thoughtfully constructed, appropriate for the task, and make full use of relevant preprocessing and model tuning strategies. Evidence of creativity and rigour in approach.	Thorough, well-justified comparison of all models using relevant accuracy metrics. Overfitting is actively analysed and mitigated using appropriate techniques (e.g. cross-validation), with strong empirical support.	Clear and critical analysis of computational efficiency, including training time, scalability, and resource usage across models. Strong empirical evidence supports conclusions.	Final pipeline is well-chosen, clearly justified against the constraints, and supported by empirical comparison. Trade-offs between models are critically evaluated, with context-sensitive reasoning, including what would change the recommendation.
B Very Good	Sound design of experiments and pipelines. The question and constraints are clear and mostly justified. Choices are mostly appropriate and well-explained, though some steps may lack optimisation or clarity. Covers all required components.	Comparison is clear and uses suitable metrics. Some attempt to analyse overfitting, with use of appropriate strategies. Evidence is generally convincing but may lack depth in places.	Good analysis of computational aspects. Covers training time and basic scalability issues. Conclusions are reasonable, though may not be fully explored.	Final pipeline is appropriate and justified against the constraints. Good comparison of alternatives with some critical insight. Minor gaps in rationale or supporting evidence.
C Good	Adequate pipeline setup. A question is stated, but the constraints are generic or only loosely linked to the models chosen. Basic design and experimentation are present, though some parts may be inconsistent, incomplete, or poorly justified.	Basic comparison of models using accuracy metrics. Some mention of overfitting, but analysis is limited or lacks justification. Mitigation strategies may be generic or underdeveloped.	Basic analysis of runtime or resource use. Limited or mostly descriptive comparison. Some claims lack supporting evidence.	Pipeline choice is reasonable but lacks strong justification; the constraints are mentioned but not tested. Trade-offs are mentioned but not critically analysed.
D Satisfactory	Minimal or inconsistent pipeline design. Constraints are missing or not used. Limited experimentation. Key stages (e.g. preprocessing or validation) are missing or flawed.	Superficial comparison. Metrics used without clear explanation or consistency. Overfitting not meaningfully addressed.	Minimal or flawed discussion of efficiency. Some reference to speed or complexity, but lacks clarity or analysis.	Pipeline is selected with little justification. Comparison is shallow or mostly descriptive.
E Weak	Pipeline setup is missing or fundamentally flawed. No evidence of meaningful experimentation or understanding of pipeline structure.	No meaningful comparison of accuracy. Overfitting issues not identified or misunderstood.	No relevant analysis of computational efficiency.	Pipeline is inappropriate or unjustified. Choice appears arbitrary or unsupported.

Presentation

Grade Band	1. Clarity of Communication (25%)	2. Depth and Accuracy of Content (25%)	3. Justification of Suggested Pipeline (25%)	4. Engagement and Q&A (25%)
A (18-22) Excellent	Clear, confident, well-paced delivery using accessible language and logical flow.	Accurate, insightful comparative analysis with clear evidence and correct terminology.	Justification is clear, well-reasoned, and demonstrates critical evaluation.	Confident, accurate responses demonstrating deep understanding.
B (15-17) Very Good	Mostly clear and confident with minor lapses in flow or clarity.	Very good analysis with strong accuracy and minor omissions.	Clear justification with good reasoning.	Clear, mostly accurate responses demonstrating good understanding.
C (12-14) Good	Generally clear but some unclear segments or pacing issues.	Good content with some gaps or inaccuracies.	Justification present but basic.	Responses mostly clear with minor inaccuracies.
D (9-11) Satisfactory	Understandable but lacks clarity or confident delivery.	Satisfactory but lacks depth; inaccuracies present.	Limited or unclear justification.	Responses basic, with some confusion or incorrect answers.
E (6-8) Weak	Unclear delivery; difficult to follow.	Weak or inaccurate content with significant omissions.	No clear justification provided.	Unable to answer questions appropriately.